Q5 has not been corrected
It is written before Q4

Department of Physics, IIT Madras

PH1020 (Physics II) **Quiz I** **20/04/2023**

Time: 8:00 – 8:50 am **Answer All questions** **Total Marks: 20**

Name	Roll. No.	Exam Hall. No.
Aryan Muralidharan	ME22B109	RMN 102

For Office Use Only

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Total marks
2	2	2	2	NA +2	0	2	3	2	15+2= sid.

Instructions:

(i) **Vectors must be indicated clearly using arrows.**

(ii) You must use only **black or blue ink** for writing the answers.

(iii) You are NOT ALLOWED to carry calculators, cell phones and other internet enabled devices.

(iv) Necessary information on the curvilinear coordinate systems as well as important vector derivatives and identities are listed on the last page.

For the questions Q1 – Q3, write the answers in the allotted box.

Q1: Write **True** or **False.** [2 × 1 = 2 *marks*]

(I) If the total charge is zero in a local distribution, the dipole moment of this distribution varies with the choice of the origin.

False

(II) $\vec{E} = ky\hat{x} - kx\hat{y}$, where k is a constant of appropriate dimension, is a valid electrostatic field.

False

$\frac{\partial E_y}{\partial x} = \frac{\partial E_x}{\partial y}$

$-K = K$ ✗

Rough

$\nabla \cdot \vec{E} = \frac{\rho_{enc}}{\epsilon_0}$

Rough

$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial & \partial_y & \partial_z \\ ky & -kx & 0 \end{vmatrix}$ $\hat{i}(-k-k) = -2k$

[2 × 1 = 2 marks]

Q2: Fill in the Blanks

(I) The inward and outward flux for a closed surface are measured to be ϕ_1 and ϕ_2 respectively. The net charge enclosed is

$\varepsilon_0(\phi_2 - \phi_1)$

(II) For a charged spherical conductor of radius R, with an irregularly shaped cavity containing a charge q, the surface charge density on the outer surface is

$q/4\pi R^2$

[2 × 1 = 2 marks]

Q3: Multiple Choice Questions

No marks will be awarded if any of the chosen option(s) is incorrect.

(I) Consider a spherical homogeneous linear dielectric with a point charge $+q$ at its center. The radius of the sphere is a and its permittivity is ϵ. Which of the following statements are correct?

(A) $\rho_b = 0$ and $\sigma_b = -\frac{q}{4\pi a^2}$

(B) ρ_b is negative and $\vec{D}$ is directed radially outwards.

(C) $\vec{\nabla} \cdot \vec{E} = \frac{\rho_b}{\epsilon_0(1+\chi_e)}$ for $0 < r < a$. χ_e is the electric susceptibility.

(D) $\vec{\nabla} \times \vec{D} = 0$ and $\vec{E} = \frac{q}{4\pi\epsilon r^2}\hat{r}$, for $0 < r < a$

D, B

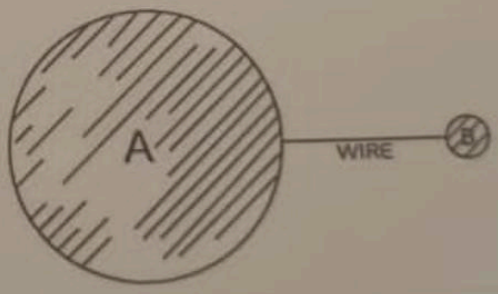
(II) There are two charged conducting spheres A and B having unequal radii and joined by a thin conducting wire as shown in the figure below. If V_A and V_B are the potentials at the surface of A and B, respectively and σ_A and σ_B are the corresponding surface charge densities, which of the following statements are true?

(A) $V_A = V_B, \sigma_A = \sigma_B$

(B) $V_A = V_B, \sigma_A < \sigma_B$

(C) $V_A \neq V_B, \sigma_A > \sigma_B$

(D) $V_A \neq V_B, \sigma_A = \sigma_B$



B

For the questions Q4 – Q9, write the answers in the attached booklet. Write your Name and Roll. No. on the first page of the booklet.

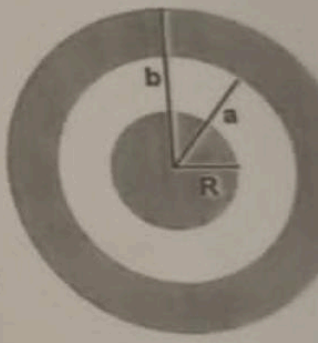
Q4 [2 marks]: Plot the potential (V) and electric field (E) as a function of the radial distance due to a solid metal sphere of unit radius and unit charge in all space. Write the values of E and V at the boundary of the sphere.

Q5 [2 marks]: A metal sphere of radius R carries a total charge Q. Calculate the force of repulsion between the northern and the southern hemispheres.

Q6 [2 marks]: Consider a homogeneous linear dielectric with permittivity ϵ. Let A be a point at the surface boundary between the dielectric and vacuum. The electric field $\vec{E}_0$ in the vacuum makes an angle α with the normal to the surface at the point A. Calculate the intensity of the electric field very close to the point A inside the dielectric.

Q7 [2 marks]: Consider a solid dielectric right-circular cone with half-angle α and height h. The cone has uniform polarization $\vec{P}$ parallel to its axis of symmetry and pointing towards its base. Calculate the total bound charge Q_b on the curved surface of the cone.

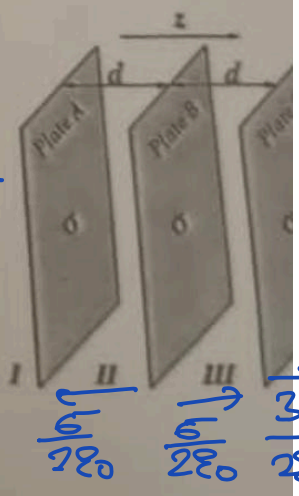
Q8 [3 marks]: A metal sphere, carrying charge q, is surrounded by a thick concentric metal shell as shown in the figure on the right. The shell carries no net charge.



(a) Calculate the potential at the center considering the reference point at infinity.

(b) If the outer shell is grounded, what will be the new potential at the center?

Q9 [3 marks]: Three large parallel plates, A, B, and C, with identical surface charge density σ are placed, with adjacent plates separated by a distance d, as shown in the figure on the right.



(a) Calculate the electric field in the regions *I*, *II*, *III*, and *IV*.

(b) What is the force per unit area on each plate?

$E_I = \sigma/\epsilon_0 \Rightarrow \frac{F}{A} = \sigma^2/\epsilon_0 = P_{III}$

$P_{II} = 0$

Q4 [2 marks]: Plot the potential (V) and electric field (E) as a function of the radial distance due to a solid metal sphere of unit radius and unit charge in all space. Write the values of E and V at the boundary of the sphere.

$E \times 4\pi r^2 = \frac{\rho \times 4/3 \pi r^3}{\varepsilon_0} \Rightarrow E = \frac{\rho r}{3\varepsilon_0}$ $\quad r < 1$

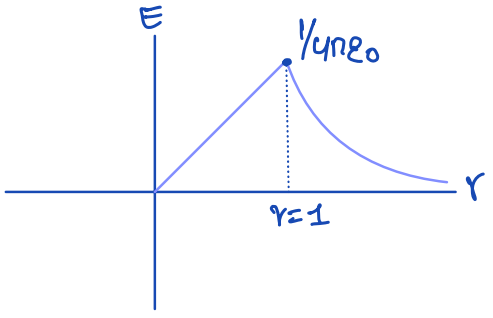
$\rho = \frac{1}{4/3\pi} = \frac{3}{4\pi}$

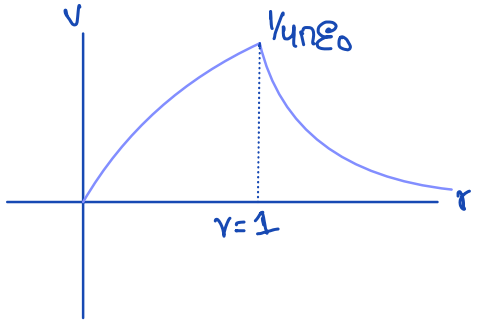
$E = \frac{r}{4\pi\varepsilon_0}$

$V = -\int_{\infty}^{r} \vec{E} \cdot d\vec{r} = -\int_{\infty}^{R} \frac{1}{4\pi\varepsilon_0 r^2} dr - \int_{R}^{r} \frac{r}{4\pi\varepsilon_0} \cdot dr = \frac{1}{4\pi\varepsilon_0} + \frac{1-r^2}{8\pi\varepsilon_0}$

$V = \frac{3-r^2}{8\pi\varepsilon_0}$

$E \times 4\pi r^2 = \frac{3/4\pi \times 4/3\pi}{\varepsilon_0} \Rightarrow E = \frac{1}{4\pi\varepsilon_0 r^2}$; $V = \frac{1}{4\pi\varepsilon_0 r}$





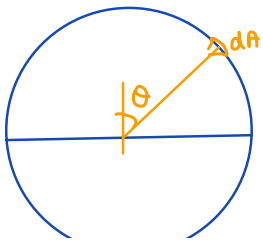
But since question said metal, charge lies outside the surface

$\Rightarrow E_{inside} = 0 \quad V_{inside} = \frac{1}{4\pi\varepsilon_0}$

$E_{outside} = \frac{1}{4\pi\varepsilon_0 r^2} \quad V_{outside} = \frac{1}{4\pi\varepsilon_0 r}$

Q5 [2 marks]: A metal sphere of radius R carries a total charge Q. Calculate the force of repulsion between the northern and the southern hemispheres.

electrostatic pressure $= \frac{\sigma^2}{2\varepsilon_0} \hat{r}$



$F = \int \sigma^2/2\varepsilon_0 \cos\theta \cdot R^2 \sin\theta \, d\theta d\phi$

$\pi/2$

$$= \frac{\sigma^2}{2\varepsilon_0} \times 2\pi \times \frac{1}{2} \int_0^{\pi} \sin 2\theta \, d\theta$$

$1/2$

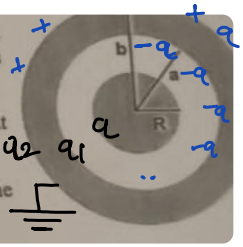
$$= \frac{Q^2}{16\pi^2\varepsilon_0 R^4} \times \pi \times \frac{1}{2}$$

$$= \frac{Q^2}{32\pi\varepsilon_0 R^4}$$

Q8 [3 marks]: A metal sphere, carrying charge q, is surrounded by a thick concentric metal shell as shown in the figure on the right. The shell carries no net charge.

(a) Calculate the potential at the center considering the reference point at infinity.

(b) If the outer shell is grounded, what will be the new potential at the center?



(a) $V = \frac{1}{4\pi\varepsilon_0}\frac{q}{R} + \frac{1}{4\pi\varepsilon_0}\frac{q}{b} - \frac{1}{4\pi\varepsilon_0}\frac{q}{a}$

(b) $\frac{kq}{b} + \frac{kq_1}{b} + \frac{kq_2}{b} = 0 \Rightarrow q_1 + q_2 = -q$

$\frac{kq}{a} + \frac{kq_1}{a} + \frac{kq_2}{b} = 0 \Rightarrow q_1 + \frac{aq_2}{b} = -q$

$\Rightarrow q_2 = 0 \quad q_1 = -q$

$$V = \frac{kq}{r} + \frac{kq_1}{q} + \frac{kq_2}{b} = \frac{kq}{R} - \frac{kq}{a}$$

INDIAN INSTITUTE OF TECHNOLOGY MADRAS

A

Roll No. | M | E | 2 | 2 | B | 1 | 0 | 9 |

Name : Aryan Muralidharan

Quiz I ✓ Quiz II/ Mid-Sem ☐ End-Semester ☐ Make-up ☐ Date : _____

Total No. of Pages ☐

Semester & Degree : ☐ Course No. ☐ Part : ☐

Question No.	1	2	3	4	5	6	7	8	9	10
Marks										

11	12	13	14	15	16	17	18	19	20	Total

Answer on both sides of the paper including the space below

Q4



R = 1 unit

Q = 1 unit

$\vec{E} = 0, \quad 0 \le r < 1$

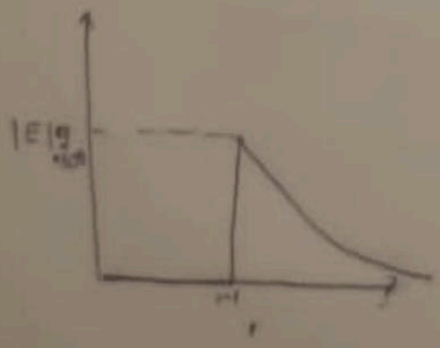
$= \frac{9 \times 10^9}{r^2} = \frac{K}{r^2} \hat{r}, \quad r \ge 1$

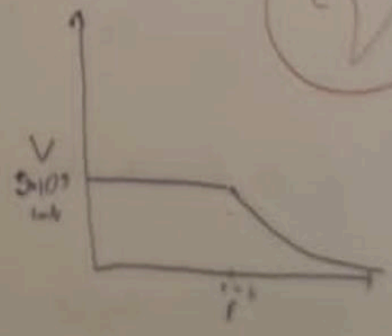
$V = 9 \times 10^9$ units $= K$ units, $0 \le r < 1$

$\frac{9 \times 10^9}{r} = \frac{K}{r}$ units, $r \ge 1$

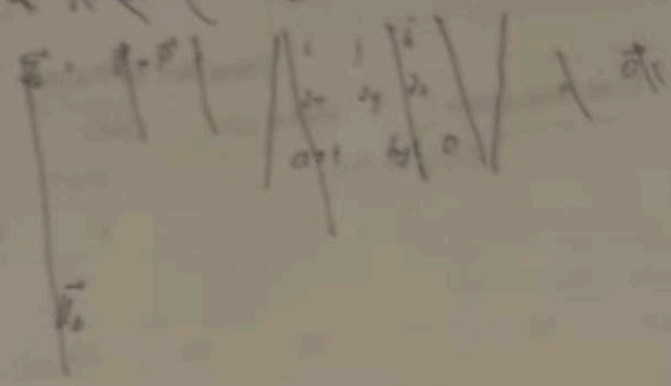
At boundary, $\vec{E} = (9 \times 10^9 \text{ units})\,\hat{r}$ (radially outwards)

$V = (9 \times 10^9$ units)





7



5. $\vec{P} = ax t\,\hat{x} + by t\,\hat{y}$

$\rho_b = -\vec{\nabla}\cdot\vec{P} = -(at + bt)$

$\sigma_b = \vec{P}\cdot\hat{n} = (ax t\,\hat{x} + by t\,\hat{y})\cdot\hat{z} = 0$

$\rho_b = -(at + bt) = -(a+b)t$

By applyg equatn of continuity :

$$\vec{\nabla}\cdot\vec{J} = -\frac{\partial \rho}{\partial t}$$

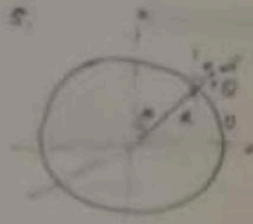
2

Here $\vec{\nabla}\cdot\vec{J}_b = (a+b)$ units $-①$

$$-\frac{\partial \rho}{\partial t} = -\frac{\partial}{\partial t}-(a+b)t = (a+b) \quad -②$$

From ① & ②: eqn of cont is satisfy ⇒ bound charge is conserd.

6.



Electrostatic pressure at a point = $\frac{\sigma^2}{2\varepsilon_0}\hat{r}$ ✓

$$\int d\vec{F} = \int_0^{2\pi}\int_0^{\pi/2} \frac{\sigma^2}{2\varepsilon_0}\cos\theta\, R^2 \sin\theta\, d\theta\, d\phi\, \hat{z} \quad (\hat{z})$$

$\int \sin\theta\cos\theta\, d\theta = \frac{1}{2}(\sin^2\theta)$

$\frac{1}{2}[1] = \frac{1}{2}$

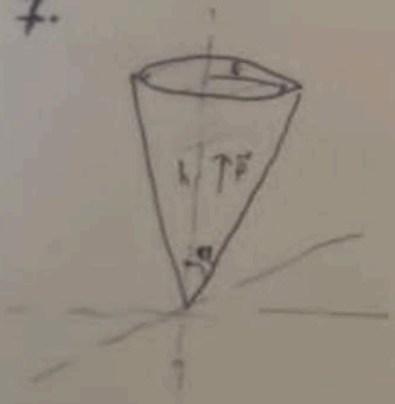
$$\vec{F} = \frac{\sigma^2 R^2}{2\cdot 2\varepsilon_0} \times 2\pi \int_0^{\pi/2} \sin 2\theta\, d\theta\, \hat{z}$$ ✓

$$\vec{F} = \frac{\sigma^2 R^2 \pi}{2\varepsilon_0}\hat{z}$$ (2)

$\sigma = \frac{Q}{4\pi R^2}$ $\therefore \vec{F} = \frac{Q^2}{16\pi^2 R^2}\cdot\frac{R^2\pi}{4\varepsilon_0}\cdot 2 = \frac{Q^2}{64\varepsilon_0\pi R^2}\cdot 2$ $(\hat{z})$

$$\vec{F} = \frac{Q^2}{32\pi R^2 \varepsilon_0}\hat{z}$$ ✓

7.



Here $\vec{P}$ is uniform

$\rho_b = -\vec{\nabla}\cdot\vec{P} = 0$ ✓

On the base of cone

$\sigma_b = \vec{P}\cdot\hat{n} = P_{unif}$ ✓

Also, $\tan\alpha = \frac{R}{h} \Rightarrow R = h\tan\alpha$.

for cone $\int \rho_b dV = \int \sigma_b dA = 0$

$\Rightarrow \int_{\text{base}} \sigma_b dA + \int_{\text{curved}} \sigma_b dA = 0$ ✓

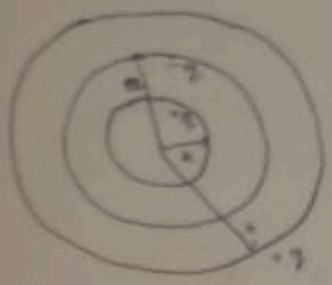
Department of Chemistry
Indian Institute of Technology Madras

17.5
+0.5
= 18

$W_{rev} = -(P_{ex} \times (\text{change in } V))$

(2) $W_{irrev} = -P(nRT\ln\frac{V_2}{V_1})$

1



By Gauss law

(a) $\vec{E} = \begin{cases} 0, & r \le a \\ \frac{kq}{r^2}\hat{r}, & a \le r \le b \\ 0, & b \le r \le c \\ \frac{kq}{r^2}\hat{r}, & r \ge c \end{cases}$

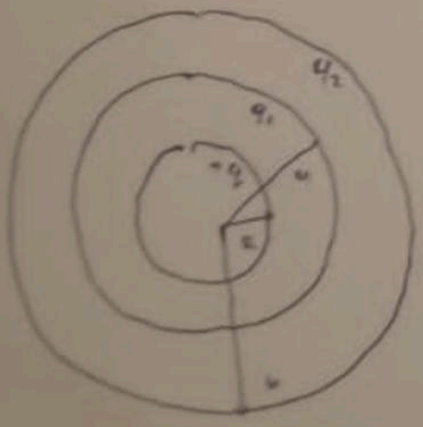
$V(\text{inside}) = -\int_{\infty}^{a} \vec{E}\cdot d\vec{r} = -\int_{\infty}^{b} \frac{kq}{r^2}dr - \int_{a}^{a} \frac{kq}{r^2}dr$

$$\boxed{V(\text{inside}) = \frac{kq}{b} + \frac{kq}{R} - \frac{kq}{a}}$$

(b) If outer shell is grounded, $V(\text{shell}) = 0$

Let new charges at the inner surface $= q_1$

outer surface $= q_2$



$V(\text{outer shell}) = 0$

$$\therefore \frac{kq}{b} + \frac{kq_1}{b} - \frac{kq_2}{b} = 0$$

$$q_1 + q_2 = -q \quad -(1)$$

$V(\text{inner shell}) = 0$

$$\frac{kq}{a} + \frac{kq_1}{a} + \frac{kq_2}{b} = 0$$

$$q + q_1 = -\frac{a}{b}q_2$$

Solving both we get:

$$\boxed{q_1 = -q}$$
$$\boxed{q_2 = 0}$$

25/05/2023

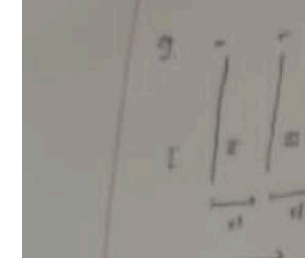
Marks: 20

xam Hall.
No.

V(new) at centre $= \frac{kq}{R} - \frac{kq}{a}$ ✓3

$V = kq\left\{\frac{1}{R} - \frac{1}{a}\right\}$

(a)



By Super-position principle:

$\vec{E}_{\text{on any pt}} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$

At region I:

$\vec{E} = \frac{\sigma}{2\epsilon_0}(-\hat{z}) + \frac{\sigma}{2\epsilon_0}(-\hat{z}) + \frac{\sigma}{2\epsilon_0}(-\hat{z})$

$\therefore \vec{E}_I = \frac{3\sigma}{2\epsilon_0}(-\hat{z})$ ✓

At II:

$\vec{E} = \frac{\sigma}{2\epsilon_0}(+\hat{z}) - 2\times\frac{\sigma}{2\epsilon_0}\hat{z}$

$\therefore \vec{E}_{II} = -\frac{\sigma}{2\epsilon_0}\hat{z}$ ✓

(2)

At III:

$\vec{E} = \frac{\sigma}{2\epsilon_0}\hat{z} + \frac{\sigma}{2\epsilon_0}\hat{z} - \frac{\sigma}{2\epsilon_0}\hat{z}$

$\therefore \vec{E}_{III} = \frac{\sigma}{2\epsilon_0}\hat{z}$ ✓

At IV:

$\vec{E}_{IV} = +\frac{3\sigma}{2\epsilon_0}\hat{z}$ ✓

(b) Force per unit area = electrostatic pressure =

$\boxed{\vec{f} = \frac{\sigma^2}{2\epsilon_0}(+\hat{z})}$ ✗

This is same for each plate as they have same value of σ